TROY BRENNAN LUNN

AI Systems & Evaluation Architect · Multi-Agent Swarm Engineer · CAD/Robotics MCP Creator · Fintech Founder
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SUMMARY & MISSION

Systems Architect, AI Evaluation Engineer, and Entrepreneur with **198,000+ lines of production code** architected and shipped across autonomous multi-agent swarms, deterministic anti-reward-hacking graders, CAD/robotics kernels, Model Context Protocol (MCP) bridges, and zero-trust fintech platforms. Daily power-user and comparative evaluator of **Gemini (Antigravity)**, **Claude (Claude Code / Cowork)**, and **Grok Build** across complex, long-horizon software and hardware engineering workflows. Former **U.S. Junior National Taekwondo Team Member (U.S. Olympic Training Center, Colorado Springs)** and **youngest All-American in the sport**—driven to build frontier AI evaluation and reasoning systems that better humanity and bring us closer to a world of unimaginable abundance.

TECHNICAL SKILLS & DEEPMIND POLARIS ALIGNMENT

- **AI Coding Agents & Evaluation Engineering:** Long-horizon SWE/CAD task design; deterministic Python/AST/B-Rep graders; anti-reward-hacking, anti-shape-spoofing & anti-sycophancy rubrics; cross-model failure analysis (**Gemini 2.5/4 in Antigravity**, **Claude Code**, **Grok Build CLI**, **OpenAI Codex**).
- **Multi-Agent Swarms & LLM Infrastructure:** Hybrid **DiffusionGemma + Autoregressive (AR) Auditor** swarms; Cloud TPUv8I KV-cache prefix sharing & JetStream continuous batching (`asyncio`); LangGraph; MCP (stdio, FastMCP SSE, JSON-RPC 2.0); on-device MediaPipe GenAI (**Gemma 3 1B-IT q8**) & OpenClaw gateways.
- **Languages, Kernels & Full-Stack Systems:** Python, Swift (`SwiftUI`), Kotlin (`Jetpack Compose`), TypeScript/JS (`React 19`, `Deno`, `Node.js`, `Electron`, `Hono`, `Three.js`), SQL (`PostgreSQL 17` / `Supabase`), OpenCASCADE / C++ CAD APIs (`build123d`, `CadQuery`, `adsk.core`/`adsk.fusion`), ROS2 `URDF`/MoveIt2 `SRDF`, EVM (`ethers.js`, Base L2).

SELECTED AI SYSTEMS, EVALUATION & ENGINEERING EXPERIENCE

Independent AI Systems Researcher & Principal Architect — Agentic Swarms, Evals & CAD/MCP Infrastructure — 2025 – Present

- **Polaris-Style Deterministic Graders & Physical Intuition Verifiers (`verify_design.py`, `tennis_match_swarm`, `layer1_agentic_loop`):** Built automated B-Rep topological complexity gates that defeat LLM **"Shape Spoofing"** (catching models that extrude 3-face cylinders named `Pinion_15T` instead of synthesizing `>20`-face involute gear teeth), discriminate NURBS spline-mesh artifacts (`<0.1 cm³`) from real collisions, and enforce `10 µm` kinematic center-distance accuracy. Formalized **Physical & Manufacturability Intuition (DFA/DFM)** into graders after finding that frontier AI lacks an intuitive grasp of the *totality* of an engineering task—designing **unbuildable "Ship-in-a-Bottle" assemblies** such as placing large bearings/motors inside a 1-piece CNC frame whose external holes are smaller than the component, or sealing a jet engine and fuel cell inside an airframe with only small intake/exhaust ports and zero split seams or access hatches.
- **Multi-Team "Tennis Match" Engineering Swarm, 1:1 Auditor Agents & DiffusionGemma Swarm (`tennis_match_swarm`, `distributed_design_swarm`, `diffusion_gemma_swarm`):** Architected a dual-orchestrator swarm where `DesignOrchestrator` (Chief Engineer) dispatches typed `DesignGoal` contracts across **7 specialized `DesignTeam` Workers** (`MDE`, `OML_Aero`, `Propulsion`, `Thermal`, `EE`, `Avionics`, `Materials`) linked by a BFS `GraphSolver`, and serves verified `.STEP` packages to `SimulationOrchestrator` across **6 verification teams** (`FEANVHTeam`, `AeroCFDTeam`, `ThermalDynamicsTeam`, `EMCSignalTeam`, `IndustrialEngineeringTeam` for DFM/DFA, `SupplyChainCostingTeam`). Paired every Worker 1-to-1 with an adversarial **`InternalAuditor` / `AuditorAgent`** catching **Reward Hacking** (deleting the `1.80 kg EngineBlock` payload), **Sycophancy**, **API Hallucinations**, and **State-Cycle Loops** (`AuditorEscalation`), plus **Cloud TPUv8I (384 MB SRAM)** `DiffusionGemmaWorker` + AR `TextAuditorAgent` KV-cache sharing.
- **Autodesk Fusion 360 State-Aware MCP Server v3.0 & Multi-LLM Bridges (`FusionStateAwareAgent`, `~18,400+ LOC`):** Built an in-process Fusion 360 add-in exposing **111 deterministic CAD/CAM/inspection MCP tools** across **Gemini/Antigravity**, **Claude Desktop**, and **Grok**, featuring thread-safe main-UI event marshalling, persistent topological entity pointers (`REFERENCE_MAP`), and every-turn bounding-box state snapshots. Engineered a custom FastMCP SSE proxy over Tailscale Funnel featuring **16 KB SSE comment padding** (defeating Cloudflare buffering) and ASGI host-header rewriting.
- **Modular Agentic CAD & Robotics Skill Suite (`.agents/skills/` — 288 files, `85,297 LOC`) & Flagship Hardware Systems:** Authored 7 comprehensive skills (`cad`, `urdf`, `srdf`, `sdf`, `render`, `step-parts`, `sendcutsend`) with a Three.js/Vite **CAD Explorer**, live **MoveIt2 WebSocket IK server**, and SendCutSend DFM preflight verifiers. Programmatically generated the **APEX-6M 6-Axis CNC Robot Arm** (`3,460 LOC`, 13 subassemblies, **142 COTS STEP parts**, 24,000 RPM ISO20 ATC spindle with twisted helical carbide cutters, 190-body Voron StealthChanger printhead, `0.0 mm` interference), a 2-stage **Hybrid SDF + OpenCASCADE B-Rep Kernel** (`1,292 LOC`) with bidirectional BFS constraint propagation (`SpatialClearanceEdge`, `LogisticalEdge`), and a 13-station analytical NACA **BWB Stealth Airframe** (`19,562 LOC` across 356 scripts).

Creator & Lead Engineer — Grok Executive Suite & ClawBridge Mobile Agent Runtime — 2026 – Present

- **Grok Executive Suite (`grok-cowork-app` — `7,690+ LOC`, 16 macOS arm64 DMG Releases `v0.1.0` → `v0.4.1`):** Built a native Apple Silicon Electron desktop workspace for Grok Build CLI featuring multi-project context injection, isolated `node-cron` scheduled execution streams, 350ms live `events.jsonl` tool-telemetry tracking, Apple Developer ID notarization, and bundled CoreGraphics/JXA macOS desktop-control MCP servers (7 stdio / 16 SSE tools).
- **ClawBridge Mobile Agent Swarm Control Surface (`com.troy.clawbridge` — `3,660+ LOC`, Android `v1.7.0`):** Built a React 19 / TypeScript 6 / Capacitor 8 Android runtime embedding an in-process Hono/Zod OpenAI-compatible gateway (`127.0.0.1:7842`) that dynamically routes between offline on-device **MediaPipe Gemma 3 1B-IT (q8)** and cloud **OpenClaw** swarm nodes provisioned via 1-click **Oracle Cloud (OCI)** `#cloud-config` + Caddy auto-TLS scripts.

Founder & Lead Systems Architect — Endless Wallet LLC (endlesswallet.io) — 2019 – Present

- Architected and shipped a **67,400+ LOC** non-custodial P2P social lending & 7-C credit evaluation (**CORE**) platform across **native iOS (`SwiftUI`, `36,555 LOC`, Build 112)**, **native Android (`Jetpack Compose`, `22,370 LOC`, v33 / SDK 36)**, **React 19 Web (`1,241 LOC`)**, and **Supabase PostgreSQL 17 (46 SQL migrations, 23 Deno Edge Functions)** by orchestrating **Gemini (Antigravity)**, **Claude**, and **Grok** across shared handoff specs.
- Engineered a **$0.00-fee Base Mainnet (Chain ID 8453) USDC settlement rail** (`c2c-pay`, `autopay-worker` with 288 `pg_cron` heartbeats/day and micro-ETH gas sponsorship); a zero-dependency **California Limited Guaranty PDF & SHA-256 e-signature engine**; and **NIST SP 800-207 Zero-Trust hardware device attestation** (custom Deno CBOR/ASN.1 DER/ECDSA P-256 Apple App Attest root CA verifier + Play Integrity + **94+ consecutive daily autonomous AI SecOps audit digests**), earning full **Plaid Production Approval**.

COMPETITIVE DISTINCTION & LEADERSHIP

- **U.S. Junior National Taekwondo Team Member & Resident Athlete, U.S. Olympic Training Center (Colorado Springs, CO):** Competed internationally at the highest level of elite athletics; recognized as the **youngest All-American in the sport** at the time of award.
- **Entrepreneur & Advanced Manufacturing Engineer, Cast Aero:** First-principles aerospace and defense manufacturing systems design.